



JUST IN TIME AND LEAN MANUFACTURING

SUMMARY

- JIT is a philosophy that was developed by the Toyota Motor Company in the mid-1970s. It has become the standard for many industries by focusing on simplicity, eliminating waste, taking a broad view of operations, visibility, and flexibility. JIT views waste as anything that does not add value.

SUMMARY CONTD

- ▶ Traditional manufacturing systems use “push” production; JIT uses “pull” production. Push systems anticipate future demand and produce in advance in order to have products in place when demand occurs. Pull systems work backwards. The last workstation in the production line requests the precise amounts of materials required.

SUMMARY CONTD

- ▶ JIT manufacturing is a coordinated production system that enables the right quantities of parts to arrive when/where they are needed. Key elements of JIT manufacturing are the pull system and kanban production, small lot sizes and quick setups, uniform plant loading, flexible resources, and streamlined layout.

SUMMARY CONTD

- ▶ TQM creates an organizational culture that defines quality as seen by the customer. The concepts of **continuous improvement and quality** at the source are integral to allowing for continual growth and the goal of identifying the causes of quality problems.

SUMMARY CONTD

- ▶ JIT considers people to be the organization's most important resource.
- ▶ JIT is equally applicable in service organizations, particularly with the push toward time-based competition and the need to cut costs.
- ▶ JIT success is dependent on inter-functional coordination and effort.

Just-in-Time (JIT)

- ▶ JIT philosophy means getting the right quantity of goods at the right place and the right time
- ▶ JIT exceeds the concept of inventory reduction; it is an all-encompassing philosophy geared to eliminate waste, anything that does not add value
- ▶ A broad JIT view – or **lean production/lean systems** - is one that encompasses the entire organization

Defining beliefs of JIT

- ▶ Everyone should have a broad view of the organization and work toward the same goal, which is serving the customer
- ▶ JIT is built on simplicity - simpler is better
- ▶ Continuous improvement – often using kaizen blitz
- ▶ Visibility – all waste must be visible to be identified and eliminated
- ▶ Flexibility - to adapt to changes in environment

ELEMENTS OF JIT



Elements of JIT Manufacturing

10

- ▶ JIT Manufacturing is a philosophy of **value-added** manufacturing
 - ▶ Inventory reduction - exposes problems
 - ▶ **Kanbans** & **pull** production systems
 - ▶ Small lots sizes & quick setups
 - ▶ Uniform plant loading
 - ▶ Flexible resources
 - ▶ Efficient facility layouts

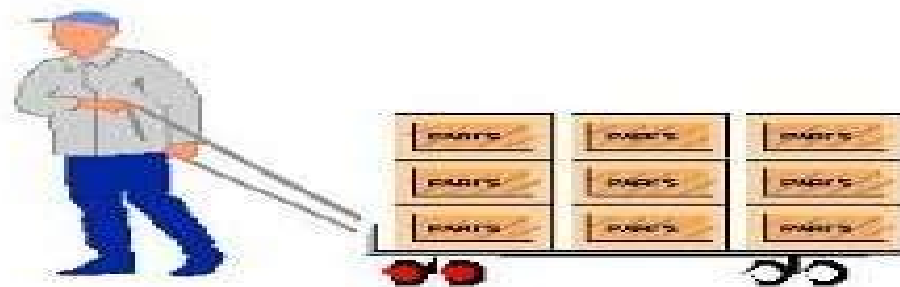
Philosophy (Cont..)

- Waste Elimination

(quality defects, inventories of all kinds, time spent to move material and time spent in setting up the machines)

- Pull System

**Make what's needed
when we need it**



- Production Precision
- Actual Consumption
- Small Lots
- Low Inventories
- Waste Reduction
- Management by Sight
- Better Communication

Elements of JIT



- Flexible Resources
- Cellular Layouts
(machines are grouped according to the process requirements)
- Kanban System
(emphasis on continual delivery)
- Small-Lot Production
- Quick Setups
- Standardization
- ability to detect when an abnormal condition)
- Total Quality Management
- Total Productive Maintenance
- Life Time Employment.

JIT - ESSENTIALS

Concept of JIT

- Consistently high quality
(Benchmarking, service design, use quality function deployment providing defect-free services)
- Uniform facility loads
(Reservation systems and differential pricing are two ways in which service providers can level the loads on their facilities)
- Standard work methods:
(analyzing work methods and standardization improvements for all employees to use)

JIT ESSENTIALS CONTINUED

- ▶ *Close supplier ties*
- ▶ (Volume services require close supplier contacts to ensure frequent, short lead time and shipments of supplier.)
- ▶ *Flexible work force*
- ▶ (multi-skilled work force)
- ▶ *Automation*
- ▶ (banks offer ATMs that provide various bank services on demand 24 hours a day)

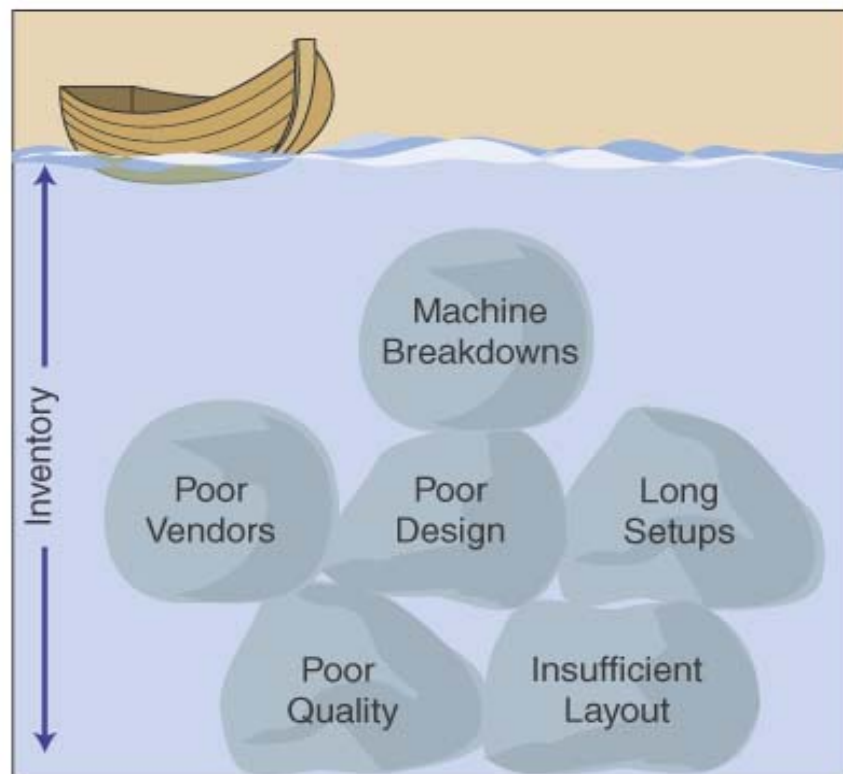
JIT TOOLS

- ▶ PREVENTIVE MAINTENANCE (Services that are highly dependent on machinery can make good use of routine preventive maintenance)
- ▶ *Pull method of material flows*
- ▶ (Services operations where tangible items are processed can utilize the pull method)
- ▶ *Product focus*
- ▶ (Managers of service operations can organize their employees and equipment)

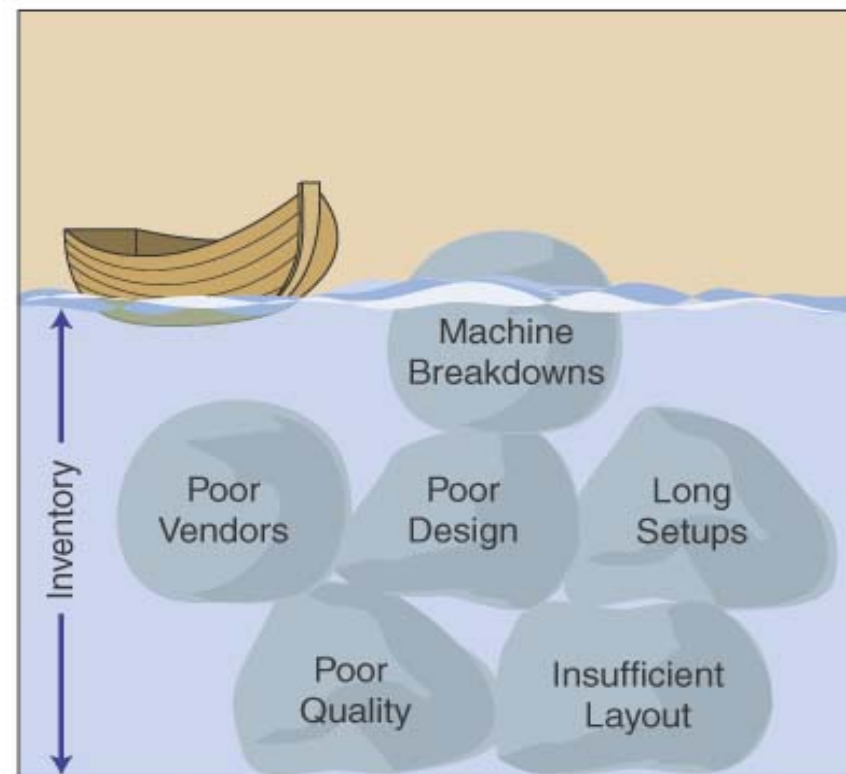
Role of Inventory Reduction

16

- Inventory = Lead Time (less is better)



(a) Inventory Hides Problems



(b) Reducing Inventory Exposes Problems

Number of Kanbans Required

17

$$N = \frac{DT + S}{C}$$

N = number of containers

D = demand rate at the withdraw station

T = lead time from supply station

C = container size

S = safety stock

Computing the Number of Kanbans: an aspirin manufacturer has converted to JIT manufacturing using kanban containers. They must determine the number of containers at the bottle filling operation which fills at a rate of 200 per hour. Each container holds 25 bottles, it takes 30 minutes to receive more bottles, safety stock is 10% of demand during LT.

Solution :

$$D = 200 \text{ bottles per hour}$$

$$T = 30 \text{ minutes} = .5 \text{ hour}$$

$$C = 25 \text{ bottles per container}$$

$$S = 0.10(\text{demand})(T)$$

$$= 0.10(200)(.5) = 10 \text{ bottles}$$

$$N = \frac{DT + S}{C} = \frac{(200)(.5) + 10}{25} = 4.4 \text{ kanban containers}$$

Question : round up or down?

Tip: round up = would provide additional slack
round down = need to make improvements in operations

Variations on Kanban Production

- ▶ Kanban boxes
 - ▶ space on factory floor for storing supplies
- ▶ Flags (signal kanban)
 - ▶ used to indicate when supplies need to be ordered
- ▶ Supplier kanbans
 - ▶ brings filled containers to point of usage in factory/picks up empty containers

Small Lot Sizes & Quick Setups

20

- ▶ Small lots mean less average inventory and shorten manufacturing lead time
- ▶ Small lots with shorter setup times increase flexibility to respond to demand changes
- ▶ Strive for single digit setups- < 10 minutes
- ▶ Setup reduction process is well-documented
 - ▶ **External setup-** do as much preparation while present job is still running; thus can be performed while machine is still running
 - ▶ **Internal setup-** simplify, eliminate, shorten steps involved with location, clamping, & adjustments; requires the machine to be stopped in order to be performed
- ▶ Ultimate goal is single unit lot sizes

Flexible Resources

- ▶ **Moveable, general purpose equipment:**
 - ▶ Portable equipment with plug in power/air
 - ▶ Drills, lathes, printer-fax-copiers, etc.
 - ▶ Capable of being setup to do many different things with minimal setup time
- ▶ **Multifunctional workers:**
 - ▶ Workers assume considerable responsibility
 - ▶ Cross-trained to perform several different duties
 - ▶ Trained to also be problem solvers

- ▶ Workstations in close physical proximity to reduce transport & movement
- ▶ Streamlined flow of material
- ▶ Often use:
 - ▶ **Cell Manufacturing**
 - ▶ Placement of dissimilar machines and equipment together to produce a family of products with similar processing requirements
 - ▶ **U-shaped lines**
 - ▶ allows material handler to quickly drop off materials & pick up finished work; flexibility

TQM

▶ **Product versus process**

- ▶ Quality in JIT is centered on building quality into the process

▶ **Quality** at the source - sequential inspection

- ▶ **Jidoka** - authority to stop line
- ▶ **Poka-yoke** - fail-safe all processes

▶ **Preventative Maintenance:**

- ▶ Regular inspections and maintenance to keep machines operational
- ▶ Costly, yes but less expensive than unexpected machine breakage.

▶ **Work environment**

- ▶ perform maintenance as part of their regular work
- ▶ Care of equipment and well-trained workers are very important.

Respect for People

The Role of Employees:

- ▶ Genuine and meaningful respect for employees
- ▶ Willingness to develop cross-functional skills
- ▶ Associates gather performance data; make production and quality decisions
- ▶ Bottom-round management
 - ▶ consensus management by committees or teams
- ▶ Quality circles
 - ▶ small volunteer teams that solve quality problems
- ▶ Everyone is responsible for Quality and preventive maintenance

Respect for People - cont'd

25

Lifetime Employment:

- ▶ Everyone feels secure/is empowered
- ▶ Everyone is responsible for quality: understand both internal and external customer needs

Respect for People cont'd

26

The Role of Management:

- ▶ Responsible for culture of mutual trust
- ▶ Serve as coaches & facilitators
 - ▶ Responsible for developing workers
 - ▶ Provide multi-functional training
 - ▶ Facilitate teamwork
- ▶ Support culture with appropriate incentive system including non-monetary

Respect for People - cont'd

27

Supplier Relationships

- ▶ Single-source suppliers
 - ▶ Can supply entire family of parts; external factory
- ▶ Build long-term relationships with small number of suppliers
 - ▶ Suppliers locate near customer
 - ▶ Fewer contracts
 - ▶ Cost and information sharing
 - ▶ Work together to certify processes

Benefits of JIT

- ▶ Reduction in inventories
- ▶ Improved quality
- ▶ Reduced space requirements
- ▶ Shorter lead times
- ▶ Lower production costs
- ▶ Increased productivity
- ▶ Increased machine utilization
- ▶ Greater flexibility

Implementing JIT

29

- ▶ Starts with a company shared vision of where it is and where it wants to go
- ▶ Management needs to create the right atmosphere
- ▶ Implementation needs a designated “Champion”

Implementing JIT – cont'd

30

Implement the sequence of seven steps:

1. Make quality improvements
2. Reorganize workplace
3. Reduce setup times
4. Reduce lot sizes & lead times
5. Implement layout changes
6. Switch to pull production
7. Develop relationship with suppliers

JIT in Services

Many JIT concepts also apply to Service companies

- Improved quality such as timeliness, service consistency, and courtesy
- Uniform facility loading to provide better service responsiveness
- Use of multifunction workers
- Reduction in cycle time
- Minimizing setup times and parallel processing
- Workplace organization

JIT & Lean Systems: How it all Fits Together

32

JIT: an overriding philosophy that affects all other business decisions

- Quality improvements
- Partnering with suppliers
- Changing job designs
- Facility layout
- Changes in production process
- Changes in inventory

JIT Across the Organization

33

JIT eliminates organizational barriers and improves communications

- ▶ Accounting changes or relies on activity-based costing
- ▶ Marketing by interfacing with the customers
- ▶ Finance approves and evaluates financial investments
- ▶ Information systems create the network of information necessary for JIT to function